Yash Dave

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Education

Stanford University June 2027
M.S. Computational and Mathematical Engineering, Data Science Track (**3.87 GPA**)
Coursework: Theoretical Statistics, AI Measurement Science, AI Reliability, Convex Optimization, Preference Learning, HCI

University of California, Berkeley May 2025
B.A. Applied Math (Economics cluster) and Computer Science (Honors); Public Policy Minor (**3.94 GPA**)
Coursework: ML*, Optimization*, Computer Vision*, Algorithms, Linear Algebra, Probability, Statistical Inference *graduate-level
Honors: Phi Beta Kappa, Outstanding GSI Award, Kevin K. Gong Memorial Scholarship, Upsilon Pi Epsilon, Dean's List

Publications & Works in Progress

Y. Dave*, S. Truong*, S. Wang†, S. Koyejo†. *The AI Evaluation Ecosystem*. arXiv:2610.09296. AI Measurement Science Workshop, COLM 2026. *equal contribution, †equal advising

M. Hardy, A. Reuel, L. Zhang, J. M. Casabianca, S. Truong, **Y. Dave**, H. Lee, B. W. Domingue, S. Koyejo. *AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems*. ICML 2026.

B. Marks, **Y. Dave**, Z. Wang, H. Chung, R. Patwa, S. Cha, M. Murphy, A. Strang. *Do Generalized-Gamma Scale Mixtures of Normals Fit Large Image Datasets?*. arXiv:2512.17038, under review.

Research Experience

Snowflake **Menlo Park, CA**
Research Intern June–August 2026
- Designed and shipped a user-in-the-loop method that mines sessions to jointly improve memory, skills, hooks, and MCP
- Built multi-turn memory evals and harness-agnostic benchmark infra for Claude Code, Codex, OpenCode, and OpenHands
- Turned real agent sessions into an internal database-engineering benchmark graded by a calibrated LLM judge

Stanford Trustworthy AI Research Lab **Stanford, CA**
Research Assistant (under Prof. Sanmi Koyejo) June 2025–present
- Building an expert-validated generative agent-based model of the AI eval ecosystem to study evaluator and policy design
- Predicting LLM agent success on unseen models and long-horizon coding tasks using explanatory item response theory

Strang Lab **Berkeley, CA**
Research Assistant (under Prof. Alexander Strang) January 2024–September 2025
- Developed a testing pipeline to evaluate realism of a Bayesian mixture model for modeling quasi-sparse image data
- Created a performant library for generating CDFs of prior distributions at scale, finding best-fit parameters, and visualization

Teaching

Course Assistant, CS 229 Machine Learning, Stanford Winter 2026
Course Assistant, CS 283 Governing Artificial Intelligence, Stanford Autumn 2025

Teaching Assistant, Data 140 Probability and Data 375 Pedagogy, UC Berkeley Fall 2024–Spring 2025
Lead Content TA, Data 100 Principles of Data Science, UC Berkeley Spring 2023–Spring 2024
- Co-authored a public debugging guide and contributed to a data science reference text

Selected Projects

Reliability and Validity of Personalized AI Benchmarks June 2026
- Built a 110-item personalized coding-agent benchmark rated by the user and a cross-lineage LLM judge panel; fit a hierarchical Bayesian Bradley-Terry-Davidson model with generalizability and decision studies to find the cheapest reliable design
- Showed LLM judges are reliable but not valid: they overstate win rates (0.80 vs. the user's 0.57), share correlated errors that do not average out, and give prediction-powered inference no label savings

Direct Preference Optimization applied to Financial Advising December 2025
- Fine-tuned a 4B-parameter model using DPO to automate SEC Marketing Rule compliance review, reducing reasoning tokens by 50% while maintaining expert-level rewrite quality
- Built a privacy-preserving synthetic data pipeline generating 943 preference pairs from 40 expert annotations using a teacher-fixer-breaker framework to train compliant ad reviewers

Beliefs about Govt. Intervention in India May 2025
- Developed a generalized linear regression model to study the relationship between local development and pro-intervention beliefs, controlling for factors like demographics, credit attribution, and government satisfaction